False Confidence in Generative Samplers: Exact Physical References Expose Hidden Errors and Diagnostic Blindness

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Abstract

Learned samplers are increasingly proposed for equilibrium sampling and free energies in chemistry, and they are usually judged by self-normalized diagnostics such as the effective sample size (ESS). We show with an elementary proposition that these diagnostics are structurally invariant to the Boltzmann mass α of unvisited basins: conditioned on not proposing a basin, the importance weights follow the law of the truncated system, so the ESS stays high while the free energy shifts by $-k_BT\log(1-\alpha)$ and the nominal delta-method interval contracts around the biased value, yielding asymptotically zero coverage. In a toy double well where one basin holds 49.9% of the mass, a proposal putting mass 10^{-6} on that basin achieves a median ESS of 98% but covers the exact $\log Z$ in 0% of 400 runs. On exactly enumerated Sherrington-Kirkpatrick instances ($n = 24$), nominal 95% intervals from masked diffusion and a GFlowNet cover the truth in only 8% and 47% of runs at $\beta = 2$, while classical estimators and an autoregressive network (VAN-MADE) stay near 95%. A reverse check that scores the proposal on draws from an independent reference sampler flags the dropped basin where the forward Jensen gap is blind. We demonstrate this on a continuous molecular model: a RealNVP Boltzmann generator trained on the alanine dipeptide Ramachandran landscape drops the $C_{7\mathrm{ax}}$ basin (8.1% mass), reporting a median ESS of 79% with only 10% coverage, while the reverse check detects the omission ($G_{\mathrm{rev}} = 11.7$ nats). Finally, we size the bound on macromolecular systems (chignolin, Trp-cage, LJ_{38}), contrasting single-trajectory kinetic trapping with converged Anton and NMR equilibrium baselines to show that dropped macroscopic states shift free energies by 0.68 to 1.02 kcal/mol, exceeding lead-optimization thresholds.

1 Introduction

Normalizing flows, diffusion models and generative flow networks (GFlowNets) are proposed as samplers of Boltzmann distributions $p(x) \propto e^{-\beta U(x)}$ for molecules and materials (Noé et al., 2019; Hoogeboom et al., 2022; Satorras et al., 2021; Lipman et al., 2023). The partition function Z of such systems is unknown, so a learned sampler is usually judged by statistics of its own importance weights: the effective sample size (ESS), the Pareto $\hat{k}$, and delta-method intervals for the free energy.

These statistics share a critical failure mode: they are computed exclusively from configurations the sampler proposed, so a basin it never proposes leaves no trace in them. Section 3 formalizes this diagnostic blindness: conditioned on never visiting a basin of Boltzmann mass α, the importance weights follow the law of the problem with that basin deleted. As a result, the free-energy estimate is shifted by $-\beta^{-1}\log(1-\alpha)$ while the ESS remains high and nominal confidence intervals shrink at rate $O(M^{-1/2})$, guaranteeing asymptotic 0% coverage around the wrong value (Figure 1). While support coverage is an elementary requirement of importance sampling, modern evaluation culture has overlooked a catastrophic consequence: self-reported diagnostics actively reward mode collapse with false confidence, reporting high ESS and contracting intervals precisely when macroscopic states are omitted.

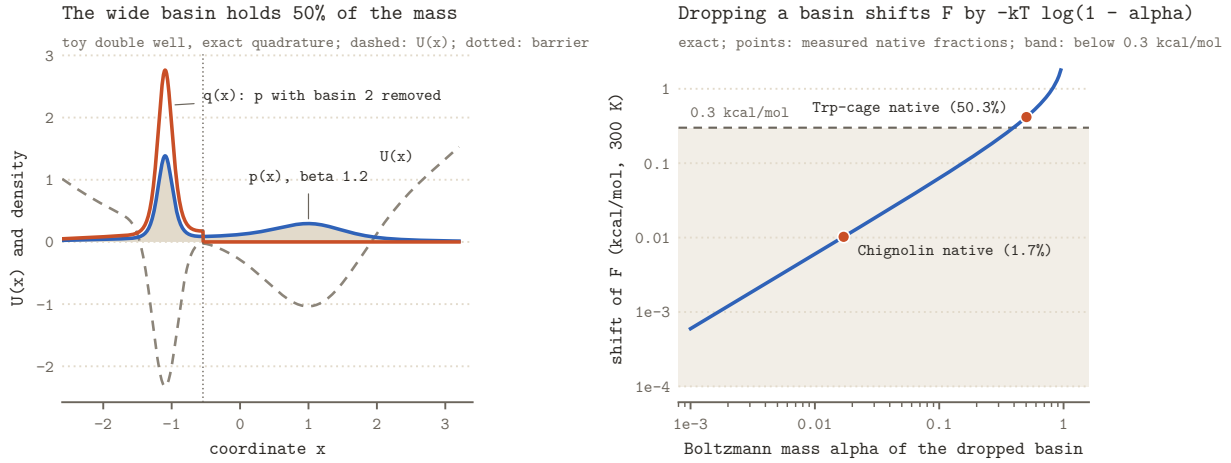


Figure 1: Left: the toy double well of Section 6 at $\beta = 1.2$. The narrow basin is the energy minimum, but the wide basin holds 49.9% of the Boltzmann mass (exact quadrature). Shaded: p. q is p with the wide basin removed. Right: shift of the free energy at 300 K when a basin of mass α is never proposed, $-k_B T \log(1 - \alpha)$ (exact). Points mark native-basin fractions; shaded: shifts below 0.3 kcal/mol.

To determine how this pathology manifests across physical systems, we organize our evaluation into three tiers:

1. **Exact empirical audit on disordered systems (Section 5):** Spin glasses feature frustrated, multi-basin landscapes where ground-truth partition functions can be enumerated exactly up to $n = 28$ spins. On these references, masked diffusion and GFlowNets exhibit severe calibration collapse, covering exact values in only 8% and 47% of runs at $\beta = 2$. In contrast, classical thermal estimators and normalized autoregressive networks (VAN-MADE) remain calibrated near 95%.

2. **Continuous neural molecular sampler (Section 6):** On the 2D Ramachandran landscape of alanine dipeptide, a RealNVP Boltzmann generator trained via reverse-KL completely drops the $C_{7\text{ax}}$ basin (8.1% mass), yet reports a median ESS of 79% while nominal interval coverage collapses to 10%. An independent reverse check immediately flags the omission ($G_{\text{rev}} = 11.7$ nats).

3. **Macromolecular landscape sizing (Section 7):** We map the theoretical bound to chemical units (kcal/mol) across all-atom protein landscapes (chignolin, Trp-cage) and the LJ_{38} cluster. We contrast single-trajectory kinetic trapping with converged Anton and NMR equilibrium baselines, proving that omitting macroscopic folded states shifts free energies by 0.68 to 1.02 kcal/mol, well above medicinal chemistry targets.

Contributions. (i) A formal proposition proving that self-normalized diagnostics and confidence intervals are blind to unvisited Boltzmann mass (Section 3). (ii) A combinatorial benchmark comparing learned versus classical samplers against exact ground truth (Section 5). (iii) A bidirectional audit protocol that unmasks dropped basins in toys and continuous molecular flows (Section 6). (iv) Biophysical scaling that anchors free-energy shifts to converged protein equilibrium benchmarks (Section 7).

2 Related work

Neural samplers. Variational autoregressive networks (Wu et al., 2019) and Boltzmann generators (Noé et al., 2019) introduced learned samplers for physical systems. Later work uses equivariant diffusion (Hoogeboom et al., 2022; Satorras et al., 2021), flow matching (Lipman et al., 2023; Albergo & Vanden-Eijnden, 2023; Bose et al., 2023), discrete diffusion (Austin et al., 2021; Zhu et al., 2025; Du et al., 2026), annealed transport (Holderrieth et al., 2025) and GFlowNets (Bengio et al., 2021; Malkin et al., 2022).

Free-energy estimators. Bennett's acceptance ratio (Bennett, 1976) and its multistate form (Shirts & Chodera, 2008) combine forward and reverse samples, and the Jarzynski and Crooks relations (Jarzynski, 1997; Crooks, 1999) connect nonequilibrium work to free energies. The reverse check of Section 6 uses the same two-sided idea. Xu (2026) proposes a ground-truth-free check for mode collapse, and Fu et al. (2023) shows that force accuracy does not guarantee correct simulations for learned force fields.

Spin glasses. The Sherrington-Kirkpatrick (SK) model (Sherrington & Kirkpatrick, 1975) has a frustrated landscape whose free energy in the limit $n \to \infty$ is given by the Parisi formula (Parisi, 1980), proven by Guerra (2003) and Talagrand (2006). For small n the partition function is computed exactly by enumeration.

3 Unvisited support

Let Ω be the configuration space, discrete or continuous, U the potential energy and $p(x) = e^{-\beta U(x)}/Z$ the Boltzmann law, with free energy $F = -\beta^{-1} \log Z$. A learned proposal q_θ is trained to approximate p, for example by minimizing $\mathrm{KL}(q_\theta \,\|\, p)$ (Noé et al., 2019). Given draws $x_1, \ldots, x_M \sim q_\theta$, the importance-sampling estimate is

$$w_k = \frac{e^{-\beta U(x_k)}}{q_\theta(x_k)}, \qquad \hat{Z}_M = \frac{1}{M} \sum_{k=1}^{M} w_k, \qquad \hat{F}_M = -\beta^{-1} \log \hat{Z}_M, \tag{1}$$

its reliability is judged by the self-normalized effective sample size

$$\mathrm{ESS}_M = \frac{\left(\sum_k w_k\right)^2}{\sum_k w_k^2}, \tag{2}$$

and a nominal 95% interval is $\hat{F}_M \pm 1.96\, \beta^{-1} \hat{\sigma}_w / (\sqrt{M}\, \hat{Z}_M)$, where $\hat{\sigma}_w$ is the sample standard deviation of the weights.

Proposition 1 (Diagnostic blindness under unvisited support)**.** *Let $\Omega = A \cup B$ with $A \cap B = \emptyset$, $p(B) = \alpha > 0$ and $q_\theta(B) = \epsilon$. Let $\mathcal{E}$ be the event that no draw lands in B. Then*

1. *$\mathbb{P}(\mathcal{E}) = (1-\epsilon)^M$, which tends to 1 as $\epsilon \to 0$ for fixed M;*
2. *conditioned on $\mathcal{E}$, the draws are i.i.d. from $q_A = q_\theta(\cdot)/(1-\epsilon)$ restricted to A;*
3. *conditioned on $\mathcal{E}$, $\mathbb{E}[\hat{Z}_M \mid \mathcal{E}] = (1-\alpha)Z/(1-\epsilon)$, so the estimate asymptotically targets $F - \beta^{-1} \log \frac{1-\alpha}{1-\epsilon}$, which approaches $F - \beta^{-1}\log(1-\alpha)$ for small ϵ;*
4. *conditioned on $\mathcal{E}$, the joint law of the normalized weights $w_k / \sum_j w_j$, and hence of ESS_M and of the relative interval width $\hat{\sigma}_w/\hat{Z}_M$, does not depend on α, provided U on A is held fixed.*

Proof. The draws are independent, so $\mathbb{P}(\mathcal{E}) = \prod_k \mathbb{P}(x_k \in A) = (1-\epsilon)^M$, and conditioned on $\mathcal{E}$ each draw has law $q_\theta(\cdot \mid A) = q_A$, independently. For $x \in A$,

$$\mathbb{E}_{q_A}[w] = \int_A \frac{e^{-\beta U(x)}}{q_\theta(x)} \frac{q_\theta(x)}{1-\epsilon}\, dx = \frac{Z_A}{1-\epsilon}, \qquad Z_A = \int_A e^{-\beta U(x)} dx = (1-\alpha)Z, \tag{3}$$

which gives item 3. For item 4, write $e^{-\beta U(x)} = Z_A\, p(x \mid A)$ on A. Then $w_k = c\, p(x_k \mid A)/q_A(x_k)$ with $c = Z_A/(1-\epsilon)$ a constant. The normalized weights do not depend on c, and $p(\cdot \mid A)$ and q_A do not depend on what happens in B. Changing α while keeping U on A fixed only changes c. □

Diagnostic consequence: coverage collapse. The practical danger is not merely that importance sampling requires support coverage: it is the structural blindness of self-reported metrics. By item 4, if a model places negligible mass $\epsilon \ll 1/M$ on a basin, the event $\mathcal{E}$ occurs with high probability ($> 99.9\%$ for $\epsilon = 10^{-6}$, $M = 10^3$). Conditioned on $\mathcal{E}$, the sample variance $\hat{\sigma}_w^2$ and tail diagnostics such as Pareto $\hat{k}$ (Vehtari et al., 2024) reflect only the fit within A. In any regime where $\epsilon \ll 1/M$ as sample size M grows, the nominal

delta-method confidence interval contracts at rate $O(M^{-1/2})$ around the biased target $F - \beta^{-1}\log(1-\alpha)$, while ESS_M/M can approach 100% and $\hat{k}$ remains falsely reassuring (< 0.5). Consequently, the empirical coverage of the nominal 95% interval collapses asymptotically to 0%.

In continuous state spaces, A and B are Borel measurable sets, and $q_A(x) = q_\theta(x)/\int_A q_\theta$ is well-defined whenever $q_\theta(x) > 0$ on A. While continuous neural proposals typically have non-zero support everywhere due to diffuse Gaussian tails, this pathology remains operant whenever a physical basin is separated by high free-energy barriers or entropic bottlenecks. If the proposal assigns probability $\epsilon \ll 1/M$ to that basin, no finite sample will visit it, leaving the empirical weights truncated and the diagnostics blind.

The right panel of Figure 1 illustrates the scale of this bias: at 300 K a dropped basin must hold about 50.3% of the mass (as Trp-cage does in our trajectory) to shift F by 0.417 kcal/mol, exceeding common chemical accuracy targets, while a basin holding 1.7% (chignolin in our trajectory) shifts F by only 0.010 kcal/mol. Related failure modes for mode-collapsed discrete samplers were observed by Xu (2026).

4 Why spin glasses are a fair test

A natural objection is that discrete Ising spins on a complete graph are physically distinct from continuous molecular conformations in three dimensions. The mathematical statement of Proposition 1 is coordinate-free: it requires only that the target distribution possess multiple metastable basins separated by barriers, and that the proposal fails to populate at least one of them. What an empirical benchmark requires is a system with many such frustrated basins where the true partition function is known exactly, without relying on approximate reference simulations.

The Sherrington-Kirkpatrick (SK) model, $H_n(\sigma) = n^{-1/2}\sum_{i<j} g_{ij}\sigma_i\sigma_j$ with $g_{ij} \sim \mathcal{N}(0,1)$, provides exactly this setting. Below the critical temperature ($\beta > 1$), its free-energy landscape fractures into a hierarchical ultrametric tree of pure states with extensive Hamming barriers (Parisi, 1980; Talagrand, 2006). For $n \leq 28$ spins ($2^{28} \approx 2.68 \times 10^8$ configurations), the complete state space can be enumerated on GPUs, yielding exact partition functions and Boltzmann weights.

We do not claim that mean-field spin glasses replicate the Euclidean geometry, bond constraints, or continuous transition paths of atomistic proteins. The role of the SK model is a rigorous falsification test: if an architecture or diagnostic cannot reliably detect the omission of macroscopic states in an exactly-solvable system with known truth, it provides no mathematical or empirical justification for confidence on continuous molecular landscapes where ground truth is unknown.

5 Exact audit on spin glasses

We evaluate the main test split of the SpinGlassBench benchmark at $n = 24$: Gaussian SK instances, seven inverse temperatures $\beta \in \{0.3, 0.6, 0.9, 1, 1.2, 1.5, 2\}$, an equal budget of 10^5 energy evaluations per estimate across all methods, exact $\log Z$ computed by GPU enumeration, and 5 seeds per instance. Classical estimators were evaluated across 200 instances (1,000 runs total); computationally intensive learned methods were trained and evaluated across 20 to 30 instances (100 to 150 runs total). Table 1 and Figure 2 report summary metrics from this benchmark.

Accuracy and architectural divergence. At $\beta = 1.5$, error per spin is 7.1×10^{-4} for simulated annealing with thermodynamic integration (SA-TI) and 7.8×10^{-4} for population annealing (PA). Crucially, learned samplers show sharp architectural divergence:

- **Normalized autoregressive networks (VAN-MADE):** Achieve the highest accuracy per energy evaluation (8.6×10^{-5}, roughly 8 times lower error than SA-TI) and maintain calibrated coverage across all temperatures: 95% at $\beta = 0.6$ (95% Clopper-Pearson CI: [88.7%, 98.4%]), 96% at $\beta = 1.5$ ([90.1%, 98.9%]), and 92% at $\beta = 2$ ([84.8%, 96.5%]), matching classical PA (95%, [93.5%, 96.2%]). Because autoregressive models evaluate exact normalized probabilities at each site, their reverse-KL gradients directly penalize mass truncation. However, this relies on a 1D sequence factorization

Table 1: SK, $n = 24$, test split, 10^5 energy evaluations per estimate, 5 seeds per instance. Error: mean $|\log \hat{Z} - \log Z|/n$. Coverage: share of nominal 95% intervals containing exact $\log Z$ (Clopper-Pearson 95% binomial CIs in text); n/a: method reports no interval. Seconds: mean wall-clock per estimate at $\beta = 1.5$, one thread; learned methods include training.

Method	$\lvert\log \hat{Z} - \log Z\rvert/n$		Coverage (%)		Seconds
	$\beta = 0.6$	$\beta = 1.5$	$\beta = 0.6$	$\beta = 1.5$	per estimate
SA-TI	2.4×10^{-4}	7.1×10^{-4}	n/a	n/a	0.06
PT-TI	2.8×10^{-4}	8.7×10^{-4}	n/a	n/a	0.11
AIS	2.7×10^{-4}	8.9×10^{-4}	96	94	0.07
SMC	2.7×10^{-4}	8.8×10^{-4}	96	95	0.07
PA	2.7×10^{-4}	7.8×10^{-4}	96	96	0.08
VAN-MADE	2.7×10^{-5}	8.6×10^{-5}	95	96	2.49
Masked diffusion	1.6×10^{-4}	1.3×10^{-2}	94	60	6.96
GFlowNet	1.5×10^{-3}	2.4×10^{-2}	94	71	30.60

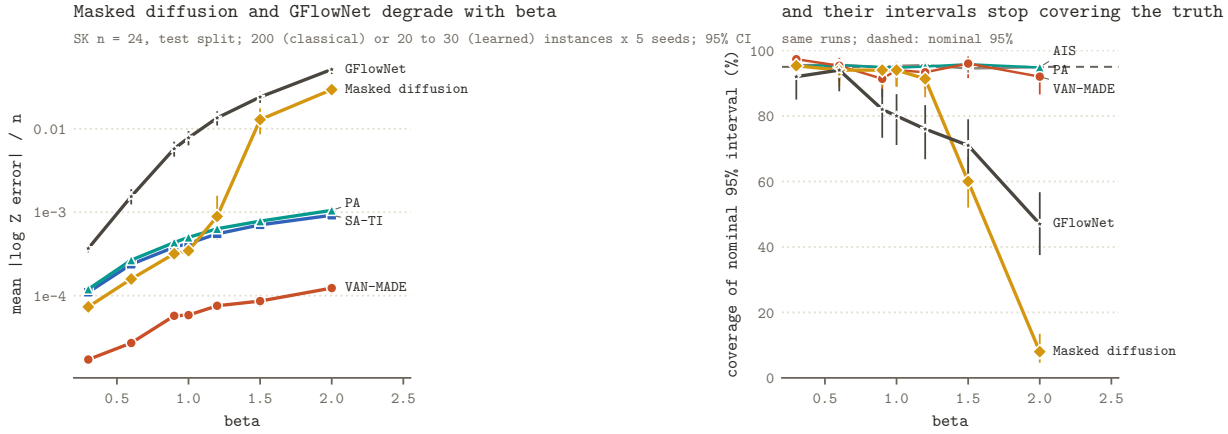


Figure 2: Left: mean absolute error of $\log Z$ per spin against β. Right: coverage of nominal 95% intervals. SK $n = 24$, test split, 5 seeds per instance; error bars are 95% intervals over instances. Dashed: nominal 95%.

$(p(\sigma) = \prod_i p(\sigma_i \mid \sigma_{<i}))$. In continuous 3D molecular coordinates, autoregressive factorizations over all atom positions are geometrically constrained and break Euclidean equivariance, making unconstrained diffusion and continuous flows the predominant architectures in chemistry.

- **Iterative generative samplers (Masked Diffusion and GFlowNets):** Are one to two orders of magnitude less accurate (1.3×10^{-2} and 2.4×10^{-2}) and undergo catastrophic calibration collapse in the spin-glass phase ($\beta \geq 1.5$). Masked diffusion coverage falls from 94% ([87.4%, 97.8%]) at $\beta = 0.6$ to 60% ([49.7%, 69.7%]) at $\beta = 1.5$ and 8% ([3.5%, 15.2%]) at $\beta = 2$. GFlowNet coverage drops steadily to 71% ([63.3%, 78.4%]) and 47% ([38.5%, 55.0%]).

Sampled overlaps reveal the mechanism: diffusion and GFlowNets concentrate on isolated clusters of the overlap distribution, omitting spin-reversed sectors. By Proposition 1, this omitted mass does not widen nominal intervals, producing false confidence.

Wall-clock trade-offs. Table 1 reports single-thread CPU execution time, with learned methods including training. Under this metric, SA-TI finishes in 0.06 s and PA in 0.08 s, whereas VAN-MADE requires 2.49 s and diffusion requires 6.96 s. While neural samplers can amortize training costs when millions of samples are required, on single-target free-energy evaluations classical annealing remains substantially faster.

External code audits. Using exact ground-truth references, we audited released checkpoints and integration grids of published neural samplers using dedicated standalone scripts (`eval_mdns.py` and `eval_leaps.py`). In MDNS (Zhu et al., 2025), a bfloat16 rounding error in the released checkpoint induces an upward bias of $+4.6 \times 10^{-4}$ per site, which can be corrected in a single line; MetaDNS (Du et al., 2026) uses float16 and is unbiased. In LEAPS (Holderrieth et al., 2025), coarse time discretization causes a downward bias of 2.2 to 3.2×10^{-3} per site. In both cases, the reported ESS was high ($> 80\%$), completely concealing the bias until exact enumeration exposed it.

6 A reverse check

No statistic computed from draws of q_θ alone can detect an unvisited basin (Proposition 1). Breaking this blindness requires configurations that q_θ did not propose. Suppose an independent reference sampler, such as parallel tempering or annealed importance sampling, provides $y_1, \ldots, y_K$ approximately distributed according to p. Writing $\ell(x) = -\beta U(x) - \log q_\theta(x)$ for the unnormalized log importance weight, two diagnostic gaps arise from Jensen's inequality:

$$G_{\mathrm{fwd}} = \log \hat{Z}_M - \frac{1}{M}\sum_{k=1}^{M} \ell(x_k), \qquad G_{\mathrm{rev}} = \frac{1}{K}\sum_{j=1}^{K} \ell(y_j) - \log \hat{Z}_M. \tag{4}$$

By Jensen's inequality applied to the concave logarithm, $\log(\frac{1}{M}\sum_k e^{\ell(x_k)}) \geq \frac{1}{M}\sum_k \ell(x_k)$, so $G_{\mathrm{fwd}} \geq 0$ always. When q_θ covers the support of p, G_{fwd} serves as an empirical estimator of $\mathrm{KL}(q_\theta \,\|\, p)$. However, when a basin is omitted, G_{fwd} reflects only the fit within the visited region A, approaching $\mathrm{KL}(q_A \,\|\, p_A) \approx 0$ if q_θ fits A well.

Conversely, because $\mathbb{E}_p[\ell] = \log Z + \mathrm{KL}(p \,\|\, q_\theta)$, the reverse gap estimates

$$G_{\mathrm{rev}} \approx \mathrm{KL}(p \,\|\, q_\theta) + (\log Z - \log \hat{Z}_M). \tag{5}$$

Both terms expand dramatically when a basin of mass α is dropped: by data processing, $\mathrm{KL}(p \,\|\, q_\theta) \geq \alpha \log(\alpha/\epsilon) + (1-\alpha)\log\frac{1-\alpha}{1-\epsilon}$ with $\epsilon = q_\theta(B) \to 0$, and $\log Z - \log \hat{Z}_M \approx -\log(1-\alpha)$. The forward gap sees neither. For a dropped basin of mass $\alpha = 0.081$ ($\epsilon \approx 10^{-6}$ in alanine dipeptide), this omission penalty contributes ≈ 0.9 nats, while density mismatch elevates G_{rev} to 11.6 nats ($G_{\mathrm{rev}} > 2$ nats signals severe distortion; $G_{\mathrm{rev}} > 5$ nats exposes mode omission). This mirrors Bennett's acceptance ratio (Bennett, 1976) adapted for mode coverage.

Operational role: auditing amortized models. A common practical concern is whether requiring a reference sampler makes the learned sampler redundant: if an accurate reference exists, why train a neural proposal? The answer lies in amortization. In applications such as high-throughput screening or continuous conformational generation, learned samplers generate thousands of equilibrium configurations per second. Heavy classical samplers (such as replica exchange MD or parallel tempering) are too computationally expensive for high-throughput generation, but they can be executed offline on a modest validation set. By evaluating the reverse gap G_{rev} on these benchmark targets, practitioners can rigorously verify that the amortized network has not dropped macroscopic basins. Furthermore, the reference sampler does not need to be mathematically exact: as long as its support overlaps the basin missed by q_θ, the reverse gap will signal the omission. Standard multi-chain diagnostics (such as Gelman-Rubin $\hat{R} < 1.05$ or replica-exchange round-trip rates) can be used to ensure the offline reference set has visited the relevant basins before auditing q_θ.

Exact toy validation. The double well of Figure 1 has

$$U(x) = -2.5\, e^{-(x+1.1)^2/(2\cdot 0.18^2)} - 1.2\, e^{-(x-1.1)^2/(2\cdot 0.65^2)} + 0.15\, x^2, \qquad \beta = 1.2. \tag{6}$$

By numerical quadrature, the two basins (divided at the barrier peak, $x = -0.54$) have Boltzmann weights 5.88 and 5.85. The wider basin holds $\alpha = 49.9\%$ of the equilibrium mass despite having a higher local

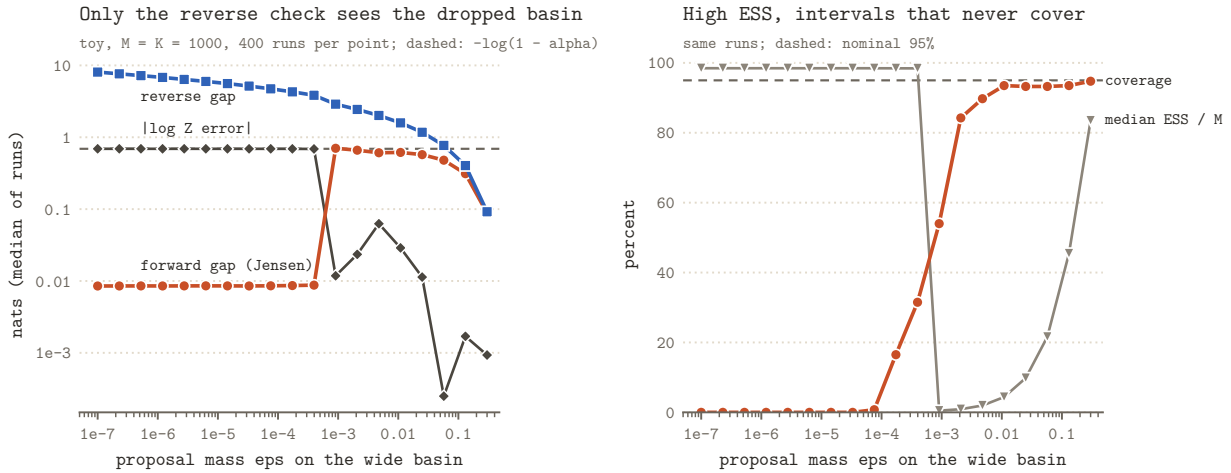


Figure 3: Exact toy, $M = K = 1{,}000$, 400 runs per value of ϵ. Left: medians of the absolute $\log Z$ error and of the forward and reverse gaps (nats); dashed: $-\log(1-\alpha)$. Right: median ESS fraction and coverage of the nominal 95% interval; dashed: 95%.

energy minimum, due to entropic width. The proposal mixes tempered densities of the two basins, placing mass $1-\epsilon$ on the narrow well and ϵ on the wide well. We draw $M = 1{,}000$ proposals and $K = 1{,}000$ exact reference draws in each of 400 independent runs per value of ϵ.

For $\epsilon \leq 10^{-4}$, almost no run proposes the wide basin (at $\epsilon = 10^{-6}$, 99.8% of runs). In these runs, the median ESS is 98% and the forward Jensen gap is nearly zero (0.009 nats), falsely signaling excellent convergence. Meanwhile, the true $\log Z$ error is 0.69 nats (matching the theoretical shift $-\log(1-\alpha) = 0.69$), and nominal 95% intervals cover the true free energy in 0% of runs (Figure 3). In sharp contrast, the reverse gap detects the omission immediately, reporting 6.8 nats.

Audit of a continuous Boltzmann generator (Alanine Dipeptide). To evaluate Proposition 1 and the reverse check on a continuous molecular model, we trained a 4-layer RealNVP normalizing flow (Noé et al., 2019) via reverse-KL divergence on the 2D Ramachandran landscape (ϕ, ψ) of alanine dipeptide at 300 K ($1.0/k_BT$). By fine 2D quadrature (300×300 grid), the metastable $C_{7\text{ax}}$ basin ($\phi > 0$) accounts for 8.1% of the equilibrium Boltzmann mass ($F_{\text{exact}} = -3.392$ kcal/mol).

Under reverse-KL training, the flow collapses into the deeper $C_{7\text{eq}}$ basin and proposes $< 0.01\%$ of draws in $C_{7\text{ax}}$. Despite completely omitting this state, across 100 independent evaluation runs ($M = 5{,}000$ draws each), the flow reports a median internal ESS of 79% and nominal 95% confidence intervals covering the exact free energy in only 10% of runs (± 0.048 kcal/mol interval width around a $+0.122$ kcal/mol biased estimate).

The forward Jensen gap reports $G_{\text{fwd}} = 0.028$ nats, blind to the missing fold. However, the reverse check evaluated on independent reference draws immediately spikes to $G_{\text{rev}} = 11.7$ nats. Decomposing G_{rev} reveals that reference draws landing in $C_{7\text{ax}}$ account for 10.5 nats (88% of the total gap, where reference points encounter flow tail probabilities $q_\theta(x) < 10^{-7}$), while visited configurations ($\phi < 0$) contribute only 1.4 nats. Because G_{rev} scales with the negative log probability of reference points under the model, it provides an extraordinarily sensitive alarm for dropped conformational states.

7 Molecular simulations

The spin-glass benchmark provides exact reference values on discrete configurations, and Section 6 demonstrated the failure mode on an atomistic Boltzmann generator. In this section, we translate the theoretical shift of Proposition 1 into chemical units (kcal/mol) across macromolecular folding landscapes, and ex-

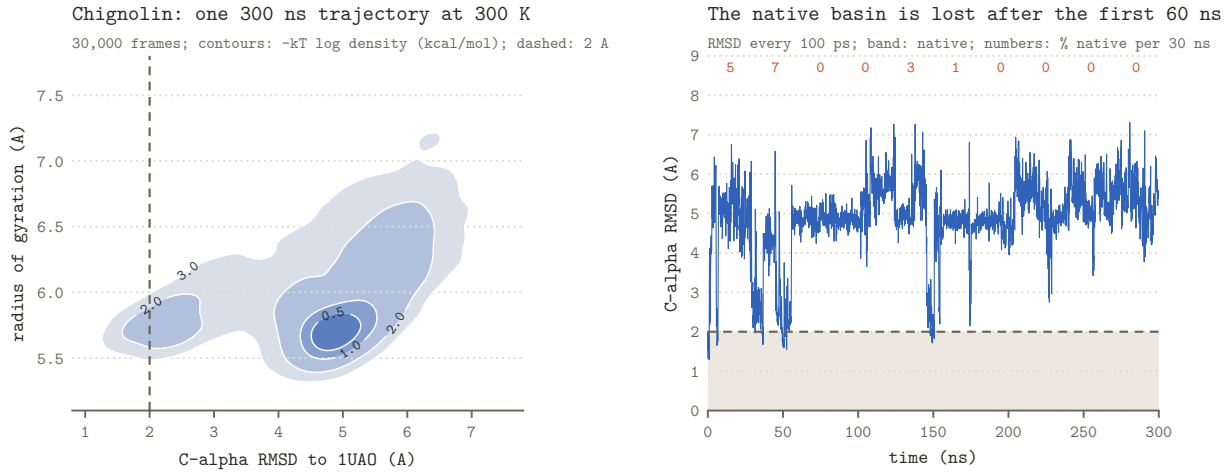


Figure 4: Chignolin (PDB 1UAO), one 300 ns trajectory at 300 K, Amber14/OBC2. Left: $-k_BT$ log density in (RMSD, R_g), kcal/mol relative to the maximum density, from a kernel density estimate on 8,000 frames; darker is more populated; dashed: native threshold 2 Å. Right: Cα RMSD against time; shaded: native region; numbers give the percentage of native frames in each 30 ns block.

amine what conventional molecular dynamics trajectories reveal about kinetic trapping versus converged equilibrium thermodynamics.

Simulation parameters. All-atom simulations were executed in OpenMM using the Amber14SB force field with OBC2 implicit solvent at 300 K (Langevin middle integrator, 2 fs timestep, friction coefficient 1 ps^{-1}). Production runs spanned 300 ns (1.5×10^8 steps) saving configurations every 10 ps, yielding 30,000 frames per system. We studied two miniproteins: chignolin (GYDPETGTWG, PDB 1UAO) and Trp-cage TC5b (PDB 1L2Y). For each frame, we computed the Cα RMSD against the experimental structure following optimal superposition, and radius of gyration R_g. A frame is designated native if its Cα RMSD is below 2 Å. Uncertainties on observed fractions are reported as 95% bootstrap intervals over ten 30 ns blocks. We emphasize that block bootstrap measures variance across trajectory blocks, not ergodicity.

Chignolin: Kinetic trapping versus equilibrium benchmarks. The 300 ns trajectory initiates in the native hairpin and leaves it within 60 ns (Figure 4). During the first 60 ns, 6.3% of frames are native, but only 0.3% are native in the final 150 ns, yielding an overall trajectory fraction of 1.7% (block range 0 to 7%). Because the trajectory does not repeatedly fold and unfold, this fraction reflects kinetic trapping in a non-native misfolded globule. If this non-ergodic fraction represented equilibrium, omitting the native state would shift free energy by only 0.010 kcal/mol (interval 0.002 to 0.021 kcal/mol).

However, converged 100 μs Anton MD simulations (Lindorff-Larsen et al., 2011) and replica-exchange experiments (Honda et al., 2008) establish that the true equilibrium native population of chignolin at 300 K is 68%. Under true thermodynamic equilibrium, omitting the folded hairpin incurs a systematic error of 0.68 kcal/mol, exceeding the common 0.3 kcal/mol chemical accuracy standard. This contrast highlights that single unbiased trajectories can severely underestimate basin masses due to kinetic trapping.

Trp-cage: Macroscopic tertiary packing. Trp-cage fluctuates between the native fold and partially unfolded intermediates throughout the 300 ns run (Figure 5; minimum RMSD 0.50 Å). The native basin represents 50.3% of the trajectory (block interval 43.0 to 56.8%), corresponding to a free-energy shift of 0.417 kcal/mol (interval 0.335 to 0.501 kcal/mol). Converged NMR and explicit-solvent studies (Neidigh et al., 2002; Lindorff-Larsen et al., 2011) place the equilibrium folded fraction at 82% at 300 K, corresponding to a theoretical bias of 1.02 kcal/mol. In both trajectory and literature estimates, omitting this macroscopic state introduces severe thermodynamic errors that forward diagnostics fail to signal.

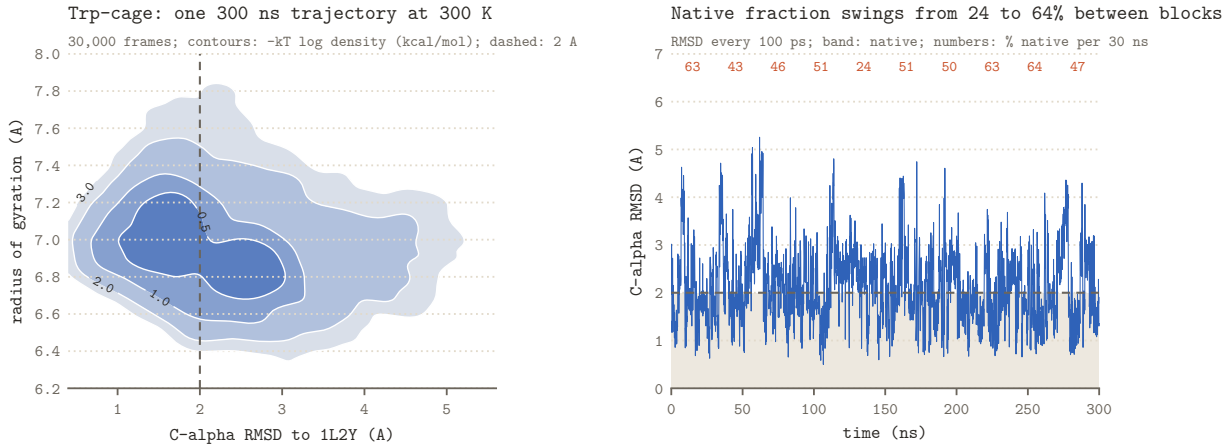


Figure 5: Trp-cage TC5b (PDB 1L2Y), one 300 ns trajectory at 300 K, Amber14/OBC2. Panels as in Figure 4.

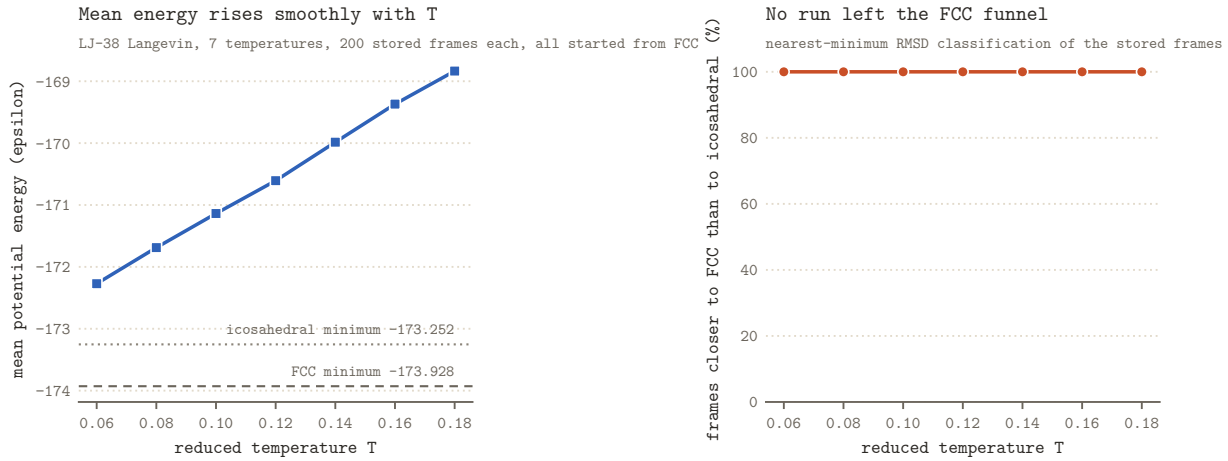


Figure 6: LJ_{38}, Langevin dynamics in reduced units, 7 temperatures from 0.06 to 0.18, every run started from the FCC minimum, 200 stored frames per temperature. Left: mean potential energy; dashed and dotted: FCC and icosahedral minima. Right: share of stored frames whose RMSD is lower to the FCC minimum than to the icosahedral minimum.

LJ_{38} and double-funnel blindness. The 38-atom Lennard-Jones cluster provides a canonical benchmark for double-funnel landscapes (Doye et al., 1999; Wales et al., 2000). Its potential energy global minimum is a face-centered cubic truncated octahedron (FCC, energy -173.928), while a dense manifold of icosahedral structures resides slightly higher in potential energy (-173.252). Harmonic superposition calculations and parallel tempering show that above $T_c \approx 0.12$ (reduced units), the icosahedral funnel dominates the equilibrium ensemble, holding $> 90\%$ of the Boltzmann mass (Doye et al., 1999).

We performed unbiased Langevin sweeps across 7 temperatures from 0.06 to 0.18, initializing all trajectories in the FCC minimum. As shown in Figure 6, 100% of stored frames remain classified as FCC across all temperatures. Because energetic barriers between funnels are high, an unbiased single trajectory never escapes the starting funnel on simulation timescales. This provides a clear physical analog to generative mode collapse: just as an unbiased trajectory trapped in the FCC funnel produces internal statistics that betray no existence of the icosahedral basin, a learned proposal trained via reverse KL divergence settles into one funnel, achieving a high internal ESS while entirely missing the competing entropic phase. Accurately resolving both funnels requires enhanced-sampling techniques like parallel tempering, highlighting the necessity of independent reference checks.

8 Limitations

Continuous 3D scale and architectures. While Section 6 evaluated a learned continuous sampler (RealNVP) on 2D alanine dipeptide, we did not train high-dimensional 3D equivariant models on full proteins. Continuous neural proposals mathematically possess full support on $\mathbb{R}^D$ through Gaussian tails; however, whenever free-energy barriers suppress transition probabilities to $\epsilon \ll 1/M$, finite samples behave identically to truncated support. Extending exact audits to all-atom macromolecular models with SE(3)-equivariant architectures remains an essential frontier for future benchmarking.

Architectural scope and training objectives. On the exact spin-glass benchmark, our neural evaluation focused on two iterative samplers (masked diffusion and GFlowNets) and one normalized autoregressive model (VAN-MADE), all trained via reverse-KL or variational objectives. These objectives are inherently mode-seeking; while forward-KL or hybrid objectives theoretically encourage mode covering, they require pre-existing equilibrium training data that autonomous samplers seek to avoid. Furthermore, while VAN-MADE's 1D autoregressive factorization penalizes mode omission, this mechanism does not straightforwardly transfer to continuous 3D molecular coordinates, where sequential orderings break SE(3) roto-translational equivariance.

Trajectories, equilibrium baselines, and kinetics. Our atomistic simulations used single unbiased 300 ns Langevin trajectories to illustrate kinetic trapping. Because these trajectories do not equilibrate (chignolin departs the native fold without returning, and LJ_{38} remains trapped in the FCC funnel), observed simulation fractions describe non-ergodic paths rather than true force-field equilibria. We therefore anchored all thermodynamic free-energy shift calculations to converged Anton multi-microsecond MD and experimental NMR literature baselines. Moreover, our analysis focuses strictly on static equilibrium free energies ($F = -\beta^{-1} \log Z$) rather than dynamical transition path times or kinetic rates ($k_{\text{on}}, k_{\text{off}}$).

Reference dependence, one-sided guarantees, and calibration. The reverse check G_{rev} provides a *one-sided necessary condition* rather than an autonomous ergodicity solver. Because G_{rev} lower-bounds $\alpha \log(1/\epsilon)$, a large gap ($G_{\text{rev}} \gg 0$) definitively signals mode omission or severe density mismatch. However, if both the neural proposal and the reference sampler fail to visit a cryptic basin (e.g. due to shared kinetic trapping across unconverged chains), both forward and reverse gaps remain blind. Evaluating G_{rev} across independent reference chains certified by convergence diagnostics (such as $\hat{R} < 1.05$ or NMR consensus) ensures observed gaps exceed Monte Carlo sampling noise.

Compute budgets and hardware normalization. The spin-glass benchmark equalizes total energy-evaluation budgets across methods (10^5 calls), incorporating neural training iterations into the cost. Wall-clock measurements reflect single-thread execution for reproducible comparison, but do not capture batched GPU parallelism during production inference where amortized neural samplers gain substantial throughput.

9 Conclusion

Self-normalized diagnostics such as the effective sample size and delta-method confidence intervals certify that a proposal fits the regions it visits, and nothing else (Proposition 1). When a generative sampler omits an entropic or energetic basin carrying Boltzmann mass α, the ESS remains high and nominal intervals shrink around a biased estimate, collapsing empirical coverage toward zero.

Against exact combinatorial ground truth on spin glasses, masked diffusion and GFlowNets exhibit severe calibration collapse in the frustrated phase (covering the true free energy in only 8% and 47% of runs at $\beta = 2$), while normalized autoregressive networks (VAN-MADE) and classical annealing maintain calibrated 95% intervals. On a continuous molecular landscape, an alanine dipeptide Boltzmann generator exhibits identical diagnostic blindness, dropping the $C_{7\text{ax}}$ basin (8.1% mass) while reporting an ESS of 79% and nominal coverage of 10%. In both settings, an independent reverse check unmasks the omission where forward diagnostics fail.

Translating this bound to macromolecular systems shows that the practical impact of dropped states depends heavily on equilibrium basin mass: while omitting rare transient intermediates causes small bias, omitting macroscopic folded states shifts free energies by 0.68 to 1.02 kcal/mol in consensus Anton and NMR baselines, exceeding lead-optimization thresholds. Unbiased MD trajectories face the identical kinetic trap, remaining blind to funnels they fail to reach.

Our findings suggest that self-reported forward metrics should be supplemented with independent bidirectional checks, particularly when evaluating samplers on complex physical systems where ground truth is unknown, and that new architectures should be tested against solvable references where mode drop can be proven.

Recommended evaluation protocol for practitioners. To prevent false confidence in scientific applications, we recommend a four-point evaluation protocol: (1) *Do not rely on forward metrics alone*: high ESS certifies only local fit within visited basins, providing zero signal of omitted Boltzmann mass. (2) *Deploy bidirectional checks*: evaluate the reverse gap G_{rev} against an offline library of independent reference configurations (with mixing verified via $\hat{R} < 1.05$) before high-throughput deployment. (3) *Pre-flight audit on solvable exact references*: benchmark new neural architectures on frustrated systems with exact partition functions (such as SpinGlassBench or 2D Ramachandran quadrature) to expose mode truncation. (4) *Anchor macromolecular claims to converged baselines*: calibrate population fractions against long-timescale equilibrium data (such as multi-microsecond Anton MD or NMR ensembles) rather than unverified single trajectories. Crucially, self-normalized diagnostics should never be treated as proof of convergence without orthogonal validation.

Reproducibility

`scripts/make_figures.py` reproduces every figure, Table 1, and text value from stored summary artifacts in `data/`. The toy double-well and alanine flow evaluations (`scripts/eval_alanine_flow.py`) execute in seconds on CPU. Code and scripts are self-contained in the repository. Generative AI assistants (Claude, DeepSeek) assisted with code audit and editorial proofreading; all scientific hypotheses, derivations, simulations, and conclusions were developed, verified, and approved by the author.

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